

Accelerating physics-informed neural networks for full waveform inversion using a hybrid quantum-classical finite-basis architecture

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Abstract

Full waveform inversion (FWI) reconstructs heterogeneous material properties from receiver data but remains computationally demanding. Physics-informed neural networks (PINNs) and their domain-decomposed variants (FBPINNs) offer a mesh-free alternative but face convergence challenges when representing complex velocity fields. We present a hybrid quantum-classical FBPINN for acoustic FWI, in which the decomposed wavefield network and the global velocity network are implemented as classical-to-quantum pipelines terminating in parameterized quantum circuits (PQCs). The PQCs are realized as differentiable JAX statevector simulators, enabling end-to-end automatic differentiation through the classical PINN, the quantum circuit, and the physics-informed loss. On a geophysical anomaly benchmark, the quantum hybrid reaches a lower L1 velocity error than the primary classical FBPINN baseline in approximately $8\times$ fewer training iterations, despite using approximately 33% fewer trainable parameters, and it outperforms all 15 classical hyperparameter variants tested. A second benchmark (checkerboard) demonstrates the generality of the inversion pipeline, confirming that the quantum hybrid architecture can recover structured spatial variations beyond the localized anomaly benchmark. Our framework is broadly applicable to wave-based inverse problems beyond geophysics, including medical ultrasound tomography and non-destructive evaluation.

Keywords: Full Waveform Inversion, Wave Propagation, Quantum Computing, Parameterized Quantum Circuit, PINNs

1 Introduction

Full waveform inversion (FWI) is a powerful technique for reconstructing material properties from recorded waveform data by exploiting the full information content of wave signals, including amplitude, phase, and frequency. Originally developed in the context of exploration geophysics for subsurface velocity imaging¹, FWI has since become central to a wide range of disciplines. In global seismology, it enables high-resolution imaging of crustal and mantle structures^{2,3}; in hydrocarbon exploration and CO₂ monitoring, it provides detailed velocity models for reservoir characterization⁴ and is increasingly applied to high-density datasets acquired via distributed acoustic sensing⁵. Beyond geophysics, FWI principles have found growing applications in medical imaging, where ultrasound computed tomography employs waveform inversion to reconstruct tissue properties for breast cancer detection and characterization^{6,7}, and in photoacoustic tomography, where wave-based inversion recovers optical absorption maps from acoustic measurements⁸. Non-destructive evaluation of engineering structures similarly leverages full waveform methods to detect defects and characterize material properties⁹. Despite its versatility across these domains,

15 FWI remains computationally demanding, requiring iterative solutions of the forward wave equation and the computation of gradients through adjoint-state methods, often at substantial cost for large-scale and high-frequency problems.

Physics-informed neural networks (PINNs) have emerged as a mesh-free alternative for solving partial differential equations (PDEs) by minimizing PDE residuals at collocation points¹⁰, approximating solutions without traditional numerical discretization. This paradigm has been extended to inverse problems, where unknown model parameters are optimized alongside the solution. Recent works have applied PINNs to wave-related problems^{11–13}, offering a fundamentally different approach to the classical adjoint-state framework. In particular, Rasht-Behesht et al.¹⁴ demonstrated the application of PINNs to FWI, achieving accurate velocity reconstruction from receiver data by jointly optimizing the wavefield and velocity networks. However, standard PINNs face well-documented challenges when applied to problems involving high-frequency wave propagation, multi-scale physics, and large computational domains. The spectral bias of neural networks, their tendency to learn low-frequency features before high-frequency ones, can severely limit convergence in wave propagation problems. To address these limitations, finite-basis physics-informed neural networks (FBPINNs) were introduced by Moseley et al.¹⁵, employing a domain decomposition strategy in which the global solution is represented as a sum of locally supported basis functions, each parameterized by an independent neural network. This approach effectively mitigates spectral bias, improves parallel scalability, and enables the solution of problems that are intractable for standard PINNs. To date, however, no work has applied the domain decomposition advantages of FBPINNs to waveform inversion.

In parallel with advances in scientific machine learning, quantum computing has attracted growing interest for its potential to accelerate computational tasks across diverse domains. Parameterized quantum circuits (PQCs), also known as variational quantum circuits (VQCs), have emerged as a near-term approach to quantum machine learning, operating within the constraints of current noisy intermediate-scale quantum (NISQ) devices¹⁶. PQCs encode classical data into quantum states through feature maps, process the information through parameterized unitary operations, and extract classical outputs via measurement. Several studies have explored the integration of quantum circuits with neural networks, forming hybrid quantum-classical architectures that leverage the expressive power of quantum Hilbert spaces while maintaining the trainability and flexibility of classical networks^{17–19}.

The intersection of quantum computing and PINNs has begun to receive attention, with recent works investigating quantum physics-informed neural networks (QPINNs) for solving differential equations²⁰. Leong et al.²¹ benchmarked hybrid quantum physics-informed neural networks (HQPINNs) for high-speed flow problems, demonstrating that these architectures offer competitive accuracy and stability with a reduced parameter cost compared to classical PINNs. While they found that pure quantum models struggle with non-harmonic features like shocks, the hybrid approach successfully mitigates these artifacts by leveraging classical sub-networks. Similarly, Berger et al.²² proposed trainable embedding quantum physics-informed neural networks (TE-QPINNs) for solving nonlinear PDEs, including the Navier-Stokes equations. Their approach achieved superior results compared to classical PINNs while maintaining the same number of parameters, highlighting the potential for more efficient optimization in high-dimensional parameter spaces. These studies suggest that quantum circuits can serve as powerful function approximators within the PINN framework, potentially offering advantages in expressivity and parameter efficiency for forward PDE problems. When extending quantum methods to geophysical inverse problems, research has thus far primarily relied on alternative computing paradigms. For instance, quantum annealing has recently been applied to seismic traveltime tomography²³ and to pre-stack and post-stack seismic inversion²⁴, with the inverse problem cast as a least-squares objective, mapped to a quadratic unconstrained binary optimization (QUBO) or Ising Hamiltonian formulation, and solved using a quantum annealer. More recently, hybrid quantum neural networks have also been explored for pre-stack and post-stack seismic inversion

within physics-informed encoder–decoder architectures²⁵. However, the application of gate-based hybrid quantum-classical neural architectures to highly nonlinear inverse problems like FWI, where both the wavefield and the velocity model are represented by learned networks and coupled through the acoustic wave equation, remains largely unexplored.

In this work, we present a hybrid quantum-classical FBPINN for acoustic FWI. The framework represents both the decomposed wavefield network and the global velocity network by classical-to-quantum pipelines: classical pre-networks map the input coordinates into low-dimensional latent spaces matched to the number of qubits, the PQCs process the encoded features through parameterized rotations and entangling gates, and classical output layers rescale the bounded circuit outputs into physical units. To isolate the architectural benefits of the quantum integration from the hardware noise of current near-term devices, the PQCs are implemented as JAX-native statevector simulators. This approach enables exact, end-to-end automatic differentiation through the classical PINN, the quantum circuit, and the physics-informed loss, avoiding the prohibitive computational overhead of parameter-shift rules required on physical hardware. This design combines the scalability advantages of FBPINN domain decomposition with the structured, bounded expressivity of PQC-based parameterizations. We evaluate the architecture on two geophysical benchmarks and compare it against a purely classical FBPINN baseline as well as 15 classical hyperparameter variants. In the following sections, we present the comparative performance of our hybrid architecture, discuss its broader implications for scientific machine learning, and detail the underlying mathematical and algorithmic methodology.

2 Results

We evaluate the proposed hybrid architecture on two wave-propagation benchmarks. The primary benchmark reconstructs a localized low-velocity anomaly from recordings acquired by a one-sided receiver array, providing a direct quantitative comparison between the quantum hybrid and a purely classical FBPINN baseline under matched FBPINN, loss, and optimizer settings. A second benchmark tests a checkerboard velocity pattern using the exact same network configuration, with only the collocation point density increased, to assess whether the proposed inversion framework can recover more spatially structured velocity variations without additional architectural tuning.

2.1 Architecture and training setup

To establish a rigorous comparison, we evaluate two distinct configurations applied to the identical FWI problem: a purely classical FBPINN baseline and our proposed hybrid quantum-classical architecture. In the classical baseline, both the decomposed wavefield network and the global velocity network are standard fully-connected tanh networks. In the hybrid model, both are replaced by classical-to-quantum pipelines that terminate in four-qubit parameterized quantum circuits (PQCs).

Beyond the network heads, both configurations share the same FBPINN subdomain decomposition strategy ($5 \times 3 \times 5$ grid with a 0.35 overlap for the wavefield), the same use of a single global velocity network, and the same physics-informed composite loss function incorporating the PDE residual, wavefield snapshot initial conditions (ICs), data misfit, and free-surface boundary constraints. Notably, the quantum hybrid model achieves its expressivity using significantly fewer trainable parameters. We note that the two configurations are not matched parameter-for-parameter: the wavefield and velocity pre-networks differ in width and depth in addition to the quantum head (Table 1), so the comparison reflects the combined classical-to-quantum substitution rather than the quantum circuit in isolation, and a parameter-matched classical control with the same backbone and a plain output head is not included here. Schematic representations of the hybrid framework (Figure 1) and the PQC topology (Figure 2) are provided alongside a summary of the network architectures in Table 1. Methodological details, including

the underlying acoustic wave equations and loss derivations, are deferred to the Methods section.

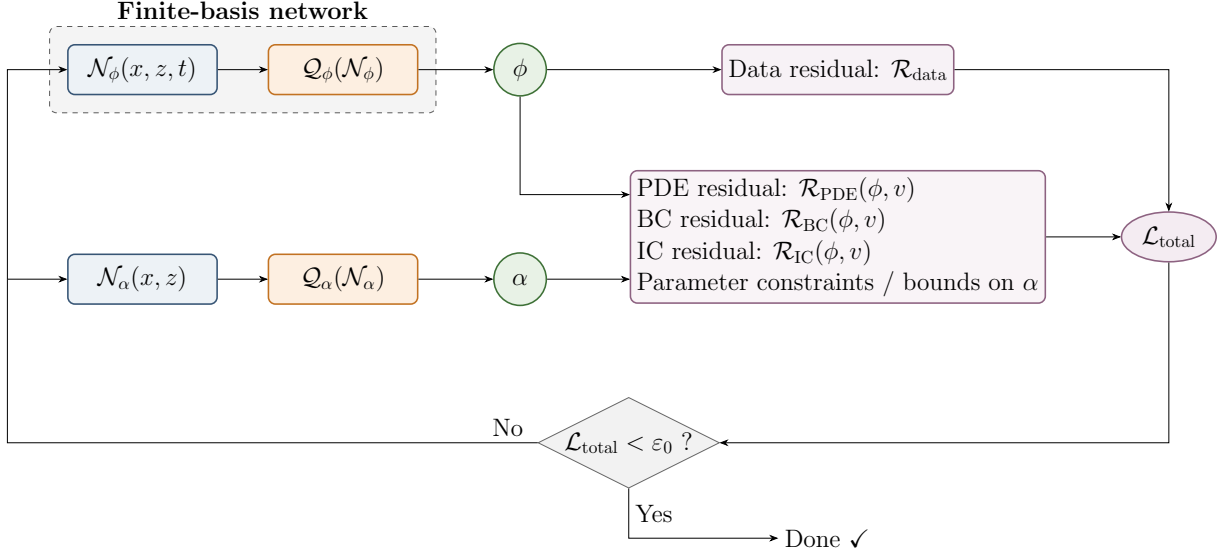


Figure 1: Schematic of the finite-basis classical-quantum hybrid architecture, combining classical neural networks and quantum circuits with physics-informed constraints. Blue boxes denote classical networks $\mathcal{N}$, orange boxes quantum circuits $\mathcal{Q}$, green circles the predicted fields ϕ and α , and purple nodes the physics- and data-based residuals contributing to $\mathcal{L}_{\text{total}}$.

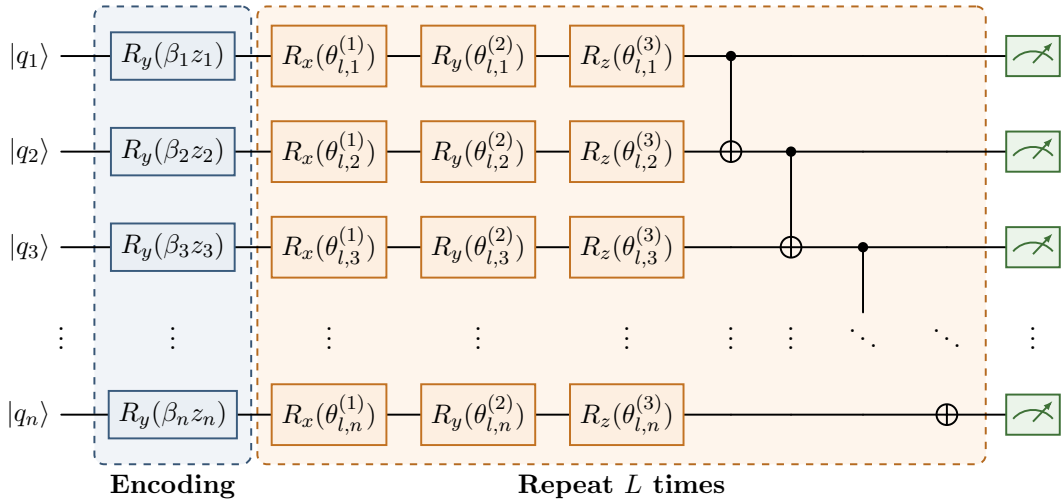


Figure 2: Multilayer PQC $\mathcal{Q}$. Each input feature z_i is angle-encoded via an $R_y(\beta_i z_i)$ rotation (blue encoding layer), where β_i is a trainable scaling parameter. Variational layers of single-qubit rotations R_x , R_y , R_z followed by a linear CNOT ladder are repeated L times (orange block). Each qubit is measured in the Pauli-Z basis (green meters), and the outputs are averaged to yield a scalar in $[-1, 1]$.

2.2 Anomaly benchmark

115 Our primary evaluation uses the velocity model established by Rasht-Behesht et al. ¹⁴, which defines a 2D acoustic domain ($1.15 \text{ km} \times 0.45 \text{ km}$) featuring a localized low-velocity anomaly ($\sim 2.5 \text{ km/s}$ embedded within a $\sim 3.0 \text{ km/s}$ background; Figure 3a). Ground-truth receiver data are generated by a spectral-element forward simulation ²⁶ driven by a Gaussian source at

Table 1: Network configurations for the anomaly benchmark. $\mathcal{Q}$: $n_q^{\times n_L}$ denotes a PQC with n_q qubits and n_L layers; classical networks use $w^{\times d}$ for d hidden layers of width w (tanh activation), and $A \rightarrow B$ denotes block A feeding block B . Subdomain decomposition ($5 \times 3 \times 5$, overlap 0.35) applies to the wavefield network ϕ ; the velocity network α is a single global network. Loss weights: $\lambda_{\text{PDE}} = 0.1$, $\lambda_{\text{IC}} = 1.0$, $\lambda_{\text{data}} = 1.0$, $\lambda_{\text{BC}} = 0.1$. Learning rate 10^{-4} . See Supplementary Table S1 for the full hyperparameter variant sweep.

Model	ϕ network $(x, z, t) \rightarrow \phi$	α network $(x, z) \rightarrow \alpha$	# Params
classical	$\mathcal{N}: 32^{\times 2} \rightarrow 16^{\times 2}$	$\mathcal{N}: 20^{\times 5}$	151,836
quantum	$\mathcal{N}: 32^{\times 2} \rightarrow \mathcal{Q}: 4^{\times 2}$	$\mathcal{N}: 20^{\times 3} \rightarrow \mathcal{Q}: 4^{\times 2}$	101,812

(0.35, 0.19) km with dominant frequency $f_0 = 20$ Hz. The inversion process is constrained by 500 ms of wavefield displacements recorded by a vertical array of 20 receivers positioned along the right boundary ($x = 1.15$ km), supplemented by two early displacement snapshots ($t_0 = 0.100$ s and $t_1 = 0.115$ s). This transmission-dominated configuration, featuring a single source and a one-sided receiver array, represents a severely ill-posed inverse problem. The limited angular illumination inherently restricts the recovery of high-wavenumber spatial components, rigorously testing the regularizing capacity of the chosen neural architecture.

Starting from a purely homogeneous initial guess (Figure 3b), both the classical and hybrid architectures successfully localize the low-velocity anomaly without suffering from the cycle-skipping artifacts that frequently plague classical adjoint-state FWI under poor starting models. As illustrated by the inverted velocity field of the quantum hybrid model (Figure 3c), the recovered anomaly closely tracks the true model’s spatial distribution and peak amplitude. The minor lateral smearing observed in the reconstruction is consistent with the theoretical limits of resolving power imposed by the single-source geometry, confirming that the network has extracted the maximum available information from the receiver data.

The clearest distinction between the classical and quantum approaches emerges in their optimization dynamics. Figure 3d tracks the L1 velocity error against training iterations, revealing structurally different convergence trajectories. The quantum hybrid model undergoes a rapid initial descent, achieving an L1 error of 2.4×10^{-2} in merely 65,000 iterations. By contrast, the classical baseline exhibits prolonged plateauing behavior, requiring 500,000 iterations to reach a comparable, albeit slightly higher, final error of 2.9×10^{-2} . This corresponds to an approximately $8\times$ reduction in the number of optimization iterations required to reach comparable or lower L1 velocity error, achieved despite the hybrid model using significantly fewer trainable parameters (101,812 vs. 151,836). We emphasize that this comparison concerns optimization iterations rather than wall-clock runtime, since the per-iteration cost differs between the classical and hybrid architectures. We also note that the traces in Figure 3d begin at different L1 errors: each model’s velocity (α) network initializes to a slightly different field about the homogeneous background, so the starting error reflects the random initialization of the differing α -network architectures rather than any quantum effect (one classical velocity-network variant initializes to essentially the same L1 as the quantum hybrid). The comparison should accordingly be read as the iteration count needed to reach a given error, not as progress from a shared starting point.

To test whether this performance gap could be explained by sub-optimal classical tuning, we conducted a systematic classical hyperparameter sweep comprising 15 distinct FBPINN variants. These baselines spanned a wide spectrum of configurations, including severely over-parameterized velocity networks (up to 60×3 hidden layers), alternative subdomain partitions ($3 \times 2 \times 3$ and $8 \times 4 \times 8$), varying learning rates, and adjusted physical loss weightings (detailed comprehensively in Supplementary Table S1). As depicted by the thin colored traces in Figure 3d, simply increasing classical capacity or tuning hyperparameters failed to bridge the gap; none of the 15 classical

configurations matched both the rapid early descent and the final L1 error achieved by the quantum hybrid within the tested training window (500,000 iterations).

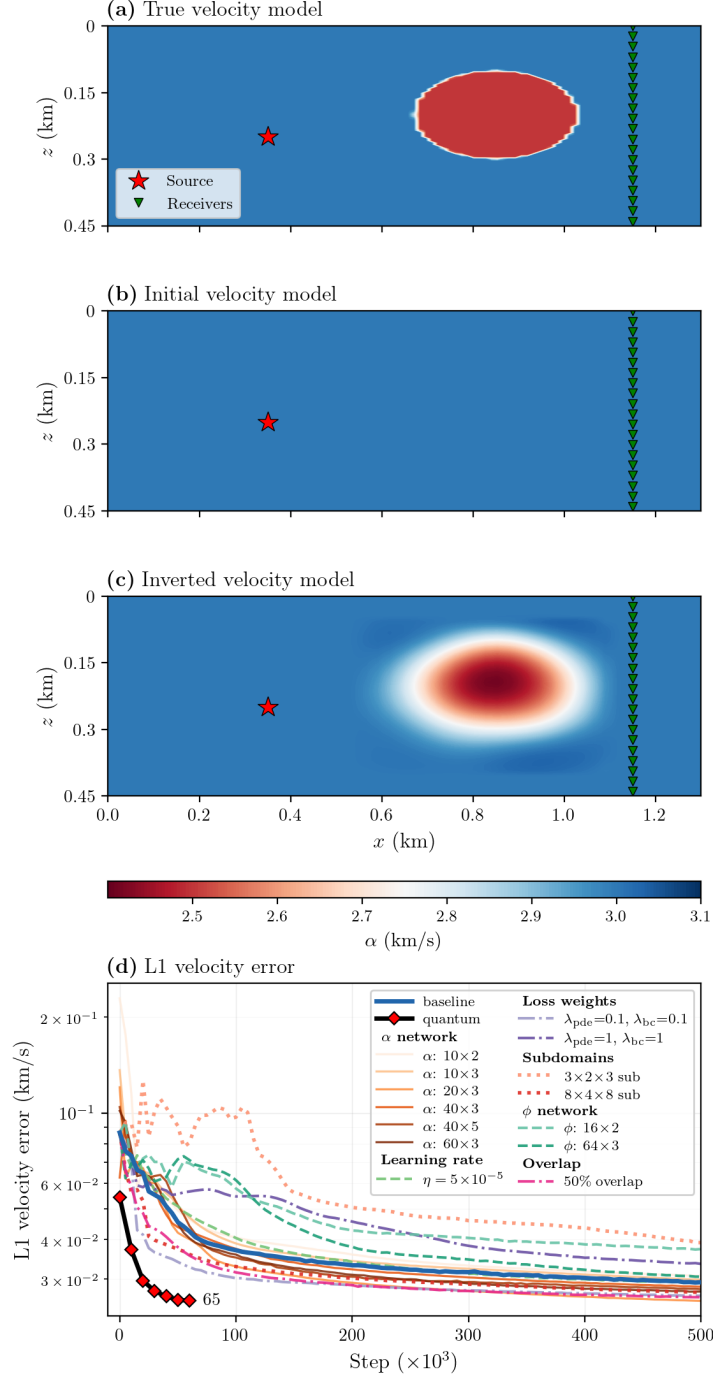


Figure 3: Anomaly benchmark inversion and convergence. (a) True velocity model used for the spectral-element forward simulation, with source (red star) and receiver array (green triangles). (b) Initial homogeneous velocity model. (c) Inverted velocity model from the quantum hybrid FBPINN at 65,000 training iterations. (d) L1 velocity error versus training iteration for the classical FBPINN baseline (blue), the quantum hybrid (red markers labeled 65k), and 14 of the 15 classical hyperparameter variants (thin colored lines). Traces begin at different initial L1 values because each velocity (α) network initializes to a different (nominally homogeneous) field; this initial spread reflects initialization rather than training.

2.3 Checkerboard benchmark

To further probe the resolution limits of the inversion pipeline, we test a checkerboard velocity model: a 3×3 grid of alternating ± 0.4 km/s perturbations on a 3.0 km/s background. The acquisition geometry is identical to the anomaly benchmark (same domain, source position, and receiver layout; Figure 3a), as is the homogeneous initial model (Figure 3b). The checkerboard forward simulation, however, uses a higher dominant source frequency, $f_0 = 60$ Hz, compared with $f_0 = 20$ Hz for the anomaly benchmark, so that the shorter wavelengths can resolve the finer spatial structure of the 3×3 checkerboard. The checkerboard experiment thus tests recovery of higher spatial-frequency velocity structure under the same limited acquisition geometry.

Both the classical and quantum hybrid FBPINNs are run with the exact same network architectures and hyperparameters established for the anomaly benchmark (Table 1). The training setup differs only in the use of a higher collocation point density, required to adequately constrain the smaller-scale spatial structures of the checkerboard pattern. Both models successfully resolve the checkerboard (Figure 4b). As expected from the single-source, one-sided receiver geometry, structural resolution degrades for blocks situated further from the receiver array. The successful inversion under this otherwise unchanged configuration confirms that the hybrid architecture generalizes beyond the anomaly benchmark without additional tuning.

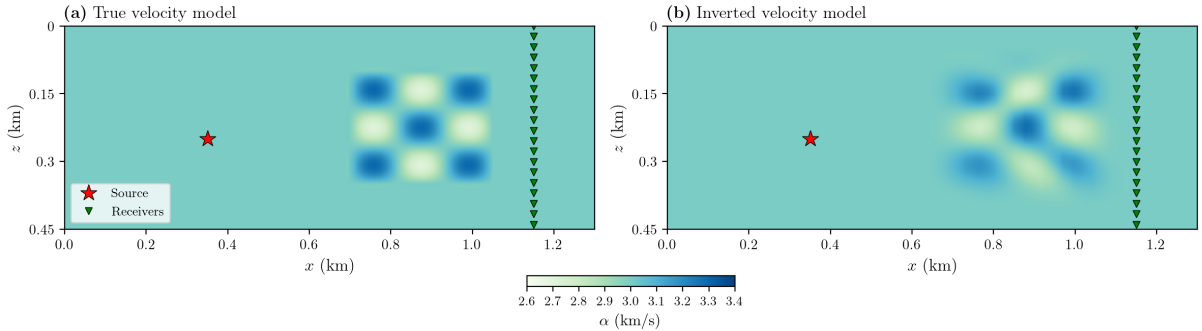


Figure 4: Checkerboard benchmark inversion. (a) True velocity model: a 3×3 checkerboard of alternating ± 0.4 km/s perturbations on a 3.0 km/s background, with source (red star) and receiver array (green triangles). (b) Inverted velocity model from the quantum hybrid FBPINN, using the identical network configuration from the anomaly benchmark with only the collocation point density increased.

3 Discussion

The quantum hybrid’s central result, a lower final L1 velocity error in $\sim 8\times$ fewer training iterations with $\sim 33\%$ fewer trainable parameters, raises the question of what in the architecture produces it. We conjecture three contributing properties of the classical-to-PQC pipeline²⁷, while noting that our experiments do not isolate their individual effects.

First, the PQC is a parameter-efficient nonlinear map. Its output is governed by only $3nL + n$ trainable parameters, yet each enters through the joint action of non-commuting rotations rather than as an independent weight, so the same angles perform both feature mixing and output generation. This shared parameterization is consistent with the hybrid model matching or exceeding the classical baseline while using $\sim 33\%$ fewer parameters. We do not invoke the dimension of the underlying Hilbert space as a source of advantage: the measured output is a single bounded scalar, and for the circuits used here ($n = 4$) it realizes only the low-order trigonometric family derived below.

Second, angle encoding endows the PQC output with a finite-frequency Fourier structure in the variables presented to the encoder. Because the PQC acts on the learned latent vector $\mathbf{z}(\mathbf{x}) = \mathcal{N}(\mathbf{x})$ rather than on the spatial coordinates directly, the relevant specialization of the result of Schuld et al.²⁸ is

$$f(\mathbf{x}; \boldsymbol{\beta}, \boldsymbol{\theta}_q) = \sum_{\boldsymbol{\omega} \in \Omega} c_{\boldsymbol{\omega}}(\boldsymbol{\theta}_q) e^{i\boldsymbol{\omega} \cdot (\boldsymbol{\beta} \odot \mathbf{z}(\mathbf{x}))}, \quad (1)$$

where the coefficients $c_{\boldsymbol{\omega}}$ are set by the trainable circuit angles and the accessible index set Ω is fixed by the encoding. For the $R_y(\beta_i z_i)$ encoding used here, each coordinate contributes only the indices $0, \pm 1$, so that for a single encoded variable

$$f(s) = A + B \cos(\beta s) + C \sin(\beta s), \quad (2)$$

with physical frequencies $0, \pm\beta_i$ that the trainable scales β_i adapt during optimization; the derivation is given in the Supplementary Note on the Fourier structure of the angle-encoded PQC. This finite-frequency structure is not unique to quantum circuits: classical Fourier-feature mappings²⁹ and sinusoidal-activation (SIREN) networks³⁰ impose a comparable trigonometric basis and are an established remedy for the spectral bias of plain tanh networks. Relative to a fixed Fourier-feature layer, the PQC differs in two respects: the encoding scales $\boldsymbol{\beta}$ are trainable, so the physical frequency grid adapts rather than being fixed a priori, and the entangling CNOT layers couple the per-coordinate features, populating cross-coordinate terms of Ω that an additive Fourier-feature layer does not produce on its own. Two caveats temper this argument: the frequency bound applies to f as a function of the learned latent $\mathbf{z}$ and not of the spatial coordinates, and our classical sweep contains only tanh architectures. A matched classical control with an explicit Fourier-feature or sinusoidal embedding is therefore the most informative comparison still missing, and we flag it as a priority below.

Third, the velocity output map hard-codes the range constraint $|\alpha - \alpha_{\text{bg}}| \leq A$. This bound is applied identically in both architectures, so it cannot by itself explain the gap; we note it only because, together with the bounded PQC output, it limits the latent signal entering the final map and may reduce large early-training excursions. Separately, the entangling CNOT layers provide a parameterization in which correlations among the latent variables are generated by the circuit dynamics rather than by stacked affine-plus-nonlinearity layers. Whether this yields a more favorable optimization trajectory, rather than a function class beyond the reach of a comparable MLP, is an empirical question we do not resolve here.

We stress that these properties bear mainly on representational capacity and parameter efficiency; none of them, on its own, predicts a faster convergence rate, so the link between the PQC parameterization and the observed iteration-count reduction remains to be established. The consistency of the improvement across the 15 classical hyperparameter variants (Figure 3d) argues against the result being solely due to an unlucky baseline, though the models also begin from different initial L1 errors set by their velocity-network initializations. We emphasize, finally, that these gains are in optimization iterations on classically simulated circuits and do not constitute a quantum computational advantage; the simulated PQC realizes a function class that is, by construction, classically representable.

Our work complements recent hybrid quantum-PINN studies in other PDE-constrained settings. Leong et al.²¹ benchmarked hybrid quantum PINNs for high-speed compressible flow, finding competitive accuracy at reduced parameter cost compared with classical PINNs. Berger et al.²² introduced trainable-embedding quantum PINNs for nonlinear PDEs including the Navier-Stokes equations, again observing efficiency gains at matched parameter count. The present work differs in two ways. First, we target a coupled forward-inverse problem (waveform inversion) rather than a pure forward PDE; the velocity field is learned jointly with the wavefield. Second, we integrate the PQC within a domain-decomposed FBPINN¹⁵, extending the PINN-based FWI formulation of Rasht-Behesht et al.¹⁴ to handle larger and higher-frequency inversion problems via overlapping subdomain networks.

Several limitations should be noted. (i) The present experiments are two-dimensional and acoustic. Extending to three-dimensional elastic inversion is the natural next step but is constrained by the $\mathcal{O}(2^n)$ memory cost of statevector simulation as qubit counts grow. Scaling to larger circuits on real hardware further raises the concern of barren plateaus (vanishingly small gradients in high-depth parameterized circuits³¹), which would require careful initialization and ansatz choice. (ii) Inversions use a single source; production FWI pipelines incorporate tens to thousands of sources with different illuminations, and PINN-based FWI in particular is known to be sensitive to loss-landscape pathologies under data sparsity³². (iii) The quantum circuit is simulated classically. On real NISQ hardware, gradients would be computed via the parameter-shift rule (two circuit evaluations per parameter), and device noise could erode the iteration-level advantage observed here.

Looking forward, the small qubit counts used here ($n = 4$, $L = 2$) are compatible with near-term quantum hardware, which suggests that genuine hardware demonstrations are feasible once noise-mitigation techniques reach sufficient maturity. Systematic studies at matched parameter counts across circuit depths and widths would clarify how the convergence advantage scales with quantum-circuit capacity. A matched classical control with an explicit Fourier-feature or sinusoidal embedding, added to the hyperparameter sweep, would in particular isolate how much of the observed advantage reflects the finite-frequency inductive bias rather than the quantum parameterization itself. Because the framework targets the general structure of physics-informed wave inversion rather than any geophysics-specific feature, we expect it to transfer directly to the broader wave-based inverse problems discussed in the Introduction, including medical ultrasound tomography, photoacoustic imaging, and non-destructive evaluation.

4 Methods

4.1 Acoustic wave equation

We consider two-dimensional acoustic wave propagation in a heterogeneous medium with spatially varying wave speed $v(\mathbf{x})$ and constant density. With no explicit source injection, the governing equation is the source-free acoustic wave equation,

$$\frac{\partial^2 u(\mathbf{x}, t)}{\partial t^2} = v^2(\mathbf{x}) \nabla^2 u(\mathbf{x}, t), \quad (3)$$

where $u(\mathbf{x}, t)$ is a scalar potential (denoted ϕ when represented by the wavefield network in the following subsections) whose spatial derivatives ($\partial u / \partial x$, $\partial u / \partial z$) give the two components of the displacement field. Wave propagation is initiated by prescribing the wavefield at two closely spaced instants, $t_0 = 0.100$ s and $t_1 = 0.115$ s, using snapshots of the displacement field taken from a spectral-element simulation; both snapshots are imposed as soft constraints in the loss function. A free-surface boundary condition (vanishing acoustic pressure, equivalently $\nabla^2 u = 0$ in this displacement-potential formulation) is enforced on the top surface of the physical domain. This formulation provides a well-posed initial-boundary-value problem without requiring an explicit source time function.

4.2 Finite-basis physics-informed neural network

PINNs¹⁰ approximate solutions of PDEs by a neural network $u_\theta(\mathbf{x})$ and train θ to minimize the PDE residual at a set of collocation points $\{\mathbf{x}_i\}_{i=1}^N$ drawn from the domain. Writing the governing PDE in residual form $\mathcal{N}[u](\mathbf{x}) = 0$, the PDE loss is

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N} \sum_{i=1}^N |\mathcal{N}[u_\theta(\mathbf{x}_i)]|^2, \quad (4)$$

with spatial and temporal derivatives computed by automatic differentiation of the network with respect to its inputs. Boundary conditions, initial conditions, and observational data are added as additional weighted loss terms (the full composite loss used for FWI is given in the Hybrid architecture subsection below). Standard PINNs, however, are known to suffer from spectral bias, a tendency to learn low-frequency features before high-frequency ones, which severely limits their effectiveness in multi-scale wave problems such as FWI, where both the velocity field and the wavefield exhibit sharp transitions and oscillatory structure.

To overcome this limitation, we adopt the FBPINN framework of Moseley et al.¹⁵. The global domain Ω is partitioned into K overlapping rectangular subdomains $\{\Omega_k\}$, each equipped with an independent neural network $u_{\theta_k}(\mathbf{x})$. The global solution is reconstructed as a partition-of-unity sum,

$$u(\mathbf{x}) = \sum_{k=1}^K w_k(\mathbf{x}) u_{\theta_k}(\mathbf{x}), \quad (5)$$

where the window functions w_k are smooth cosine bumps supported on Ω_k with $\sum_k w_k(\mathbf{x}) = 1$ for all $\mathbf{x} \in \Omega$. Each local network only needs to represent the solution's behavior on its own subdomain, with two consequences: (i) the local frequency content each network must learn is bounded by the subdomain size, alleviating spectral bias; and (ii) training is embarrassingly parallel across subdomains, enabling vectorized batched evaluation in JAX. The physics-informed loss is evaluated on the reconstructed global solution, so gradients flow through the window-weighted sum back to the individual subdomain networks.

For the wavefield $\phi(x, z, t)$ we use a three-dimensional subdomain decomposition (n_x, n_z, n_t) , $5 \times 3 \times 5$ in the anomaly baseline with subdomain overlap 0.35. The velocity network $\alpha(x, z)$ is not decomposed; it is a single global network, since the velocity field typically varies more smoothly than the wavefield and does not require local representation.

4.3 Parameterized quantum circuits

Basic background on qubits, quantum states, and gates is provided in the Supplementary Information. Here we use only the quantum ingredients required by the hybrid architecture. An n -qubit register is represented by a normalized statevector in $\mathbb{C}^{2^n}$. The circuit uses parameterized single-qubit rotations

$$R_k(\theta) = e^{-i\theta \sigma_k/2}, \quad k \in \{x, y, z\}, \quad (6)$$

where σ_k are the Pauli matrices, together with controlled-NOT (CNOT) gates to introduce entanglement between neighboring qubits.

4.3.1 PQC ansatz

A PQC¹⁸ applies a trainable unitary $U(\boldsymbol{\theta}_q)$ to the initial state $|0\rangle^{\otimes n}$ and returns a scalar expectation value of a Hermitian observable, yielding a smooth, differentiable function of the circuit parameters $\boldsymbol{\theta}_q$. Its expressibility depends on the circuit depth, qubit connectivity, and gate set³³.

The PQC employed in this work (Figure 2) operates on n qubits with L variational layers. In the data-embedding layer, starting from $|0\rangle^{\otimes n}$, each qubit i receives an $R_y(\beta_i z_i)$ rotation (angle encoding³⁴) that encodes the i -th component of the input vector $\mathbf{z} \in \mathbb{R}^n$, scaled by a trainable basis parameter β_i :

$$|\psi_{\text{enc}}(\mathbf{z}; \boldsymbol{\beta})\rangle = \bigotimes_{i=0}^{n-1} R_y(\beta_i z_i) |0\rangle. \quad (7)$$

This is followed by L variational layers. Each layer $l = 1, \dots, L$ applies a general single-qubit rotation $R_z(\theta_{l,i}^{(3)}) R_y(\theta_{l,i}^{(2)}) R_x(\theta_{l,i}^{(1)})$ to every qubit i , introducing $3n$ trainable parameters per layer, and then applies a linear CNOT ladder $\text{CNOT}(i, i+1)$ for $i = 0, \dots, n-2$ to entangle neighboring

qubits. The total number of trainable gate parameters is $3nL$, plus n basis parameters from the embedding layer. In the measurement layer, the Pauli- Z expectation values $\langle Z_i \rangle$ on each qubit are averaged to produce a single scalar output

$$f(\mathbf{z}; \boldsymbol{\beta}, \boldsymbol{\theta}_q) = \frac{1}{n} \sum_{i=0}^{n-1} \langle \psi(\mathbf{z}; \boldsymbol{\beta}, \boldsymbol{\theta}_q) | Z_i | \psi(\mathbf{z}; \boldsymbol{\beta}, \boldsymbol{\theta}_q) \rangle \in [-1, 1]. \quad (8)$$

4.4 JAX-based statevector simulator

We implement the PQC as a lightweight statevector simulator in JAX³⁵ rather than relying on an external framework such as PennyLane³⁶ or Qiskit³⁷. The simulator represents the quantum state as a 2^n -dimensional complex vector and applies gate operations via tensor contractions: single-qubit rotation gates (R_x , R_y , R_z) are constructed as 2×2 unitary matrices and applied with `jnp.tensordot` on the reshaped state tensor, and CNOT gates are realized through index permutation and conditional flipping on the control-target subspace. Expectation values are computed analytically from the statevector amplitudes, without stochastic sampling, as in PennyLane’s `default.qubit` backend.

On near-term quantum hardware, gradients of PQC outputs with respect to gate parameters are typically computed via the parameter-shift rule^{38,39}, which evaluates the circuit at two shifted parameter values per gate and therefore requires $2p$ circuit evaluations per gradient for a circuit with p parameters. In classical statevector simulation, each gate application is already a standard differentiable operation, so embedding the circuit in JAX offers two concrete advantages. First, `jax.grad` and `jax.jvp` backpropagate through the full hybrid pipeline (classical PINN layers, quantum circuit, and loss function) without parameter-shift rules or cross-framework bridging. Second, the entire training loop can be compiled into a single optimized XLA program via `jax.jit`, eliminating Python-level dispatch and framework-bridging overhead. The trade-off is that statevector simulation scales as $\mathcal{O}(2^n)$ in memory and $\mathcal{O}(Ln2^n)$ in computation, limiting the approach to moderate qubit counts; for the circuit sizes used here ($n \leq 8$, $L \leq 6$), this cost is negligible compared with the classical network and loss evaluation. We verify numerical correctness by comparing parameter gradients $\partial f / \partial \boldsymbol{\theta}_q$ against PennyLane’s `default.qubit` backend³⁶ across eight circuit configurations (2–8 qubits, 1–6 layers); the maximum absolute deviation is below 2×10^{-7} , consistent with `float32` machine precision (Supplementary Figure S1).

4.5 Hybrid quantum-classical architecture

The hybrid architecture (Figure 1) couples two neural networks with physics-informed constraints: a wavefield network $\phi(x, z, t)$, decomposed across subdomains, and a velocity network $\alpha(x, z)$, which is a single global network. In the classical baseline, both networks are fully-connected tanh networks. In the quantum hybrid, both networks are classical-to-quantum pipelines.

Defining the latent vector

$$\mathbf{z}(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\theta}_c), \quad (9)$$

each classical-to-quantum pipeline has the form

$$g(\mathbf{x}) = w_{\text{out}} \mathcal{Q}(\mathbf{z}(\mathbf{x}); \boldsymbol{\beta}, \boldsymbol{\theta}_q) + b_{\text{out}}, \quad (10)$$

where $\mathcal{N}(\mathbf{x}; \boldsymbol{\theta}_c)$ is a classical fully-connected tanh sub-network with parameters $\boldsymbol{\theta}_c$, $\mathcal{Q}(\cdot; \boldsymbol{\beta}, \boldsymbol{\theta}_q)$ is the PQC with encoding scales $\boldsymbol{\beta}$ and variational circuit parameters $\boldsymbol{\theta}_q$ (returning a scalar in $[-1, 1]$), and $(w_{\text{out}}, b_{\text{out}})$ are learnable output weights. For the anomaly benchmark experiments we use $n = 4$ qubits and $L = 2$ variational layers in both networks. The velocity network output is further mapped to physical velocity units via

$$\alpha(x, z) = \alpha_{\text{bg}} + A \tanh(g(x, z)) \cdot m(x, z), \quad (11)$$

with background velocity $\alpha_{\text{bg}} = 3.0 \text{ km/s}$, amplitude $A = 2.0 \text{ km/s}$, and a smooth spatial mask $m(x, z) \in [0, 1]$ (the product of four logistic cutoffs) that confines the velocity perturbation to the interior of the physical domain. This construction hard-codes physical bounds ($|\alpha - \alpha_{\text{bg}}| \leq A$) and zero perturbation at the domain edges, acting as a mild regularization of the inverse problem.

The full training objective is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{data}} \mathcal{L}_{\text{data}}(\phi) + \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}}(\phi, v) + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}}(\phi) + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}}(\phi), \quad (12)$$

where $\mathcal{L}_{\text{data}}$ is the data misfit between the predicted and observed displacement fields at the receiver locations, $\mathcal{L}_{\text{PDE}}$ enforces the source-free acoustic wave equation (3) via automatic differentiation, and $\mathcal{L}_{\text{IC}}, \mathcal{L}_{\text{BC}}$ are the snapshot-based initial-condition and free-surface boundary-condition constraints. All parameters (classical-network weights θ_c , embedding scales β , variational circuit angles θ_q , and output scalings) are optimized jointly by Adam gradient descent through the full differentiable pipeline.

4.6 Training setup

All training uses the Adam optimizer at learning rate 10^{-4} (baseline; see Supplementary Table S1 for variants). Each training step uses a collocation batch of shape $(n_x, n_z, n_t) = (40, 40, 25)$ across the subdomains (40,000 PDE collocation points in total), together with $40 \times 40 = 1,600$ points per displacement snapshot for the two initial-condition constraints, $\sim 1,000$ receiver observations (20 receivers subsampled every 100 simulation timesteps), and 5,000 randomly sampled boundary-condition points on the top surface. The composite loss uses fixed weights $\lambda_{\text{PDE}} = 0.1$, $\lambda_{\text{IC}} = 1.0$ (applied equally to the two snapshot constraints), $\lambda_{\text{data}} = 1.0$ (receiver data), and $\lambda_{\text{BC}} = 0.1$; the per-component loss decomposition for the classical baseline over training is shown in Supplementary Figure S2. The classical baseline is trained for 500,000 iterations; the quantum hybrid is trained for 65,000 iterations, at which point it has already surpassed the classical baseline’s 500,000-step L1 velocity error.

Training uses JAX’s just-in-time compilation and vectorized batching over subdomains to compile the update step into a single XLA program. The complete update step (classical-network forward/backward passes, PQC statevector evolution, Jacobian computation by automatic differentiation, and the composite loss) executes as a single GPU kernel. For the quantum hybrid, the statevector footprint is negligible: the 4-qubit PQC requires only $2^4 = 16$ complex amplitudes per forward pass, so memory cost is dominated by the classical wavefield subdomain networks. Spatial coordinates were scaled by $L_x = L_z = 3 \text{ km}$ in all training computations.

5 Conclusions

We introduced a hybrid quantum-classical FBPINN for acoustic FWI, in which the decomposed wavefield network and the global velocity network are represented by PQC-based classical-to-quantum pipelines within a domain-decomposed physics-informed framework, implemented as an end-to-end differentiable JAX program that avoids parameter-shift-rule overhead and jointly optimizes the classical PINN parameters and quantum circuit angles. On the anomaly benchmark, the hybrid model reaches a comparable or lower L1 velocity error in $\sim 8\times$ fewer training iterations than the primary classical FBPINN baseline, despite using $\sim 33\%$ fewer trainable parameters, and it outperforms all 15 classical hyperparameter variants tested within the same 500,000-iteration window. We emphasize that these gains are in optimization iteration count on classically simulated circuits and do not constitute a claim of quantum computational advantage, and that the mechanism behind the iteration-count reduction remains to be established. Key limitations include reliance on classical statevector simulation, a two-dimensional acoustic setup, a single-source configuration, and a comparison that is not parameter-matched and spans only tanh classical architectures; a matched classical control with an explicit Fourier-feature or sinusoidal

embedding is the most informative experiment still needed to isolate how much of the advantage reflects a finite-frequency inductive bias rather than the quantum parameterization. Extension to elastic, three-dimensional, and multi-source inversion, and validation on near-term quantum hardware, are the natural next steps. Because the framework targets the general structure of physics-informed wave inversion rather than any geophysics-specific feature, we expect it to transfer to other wave-based inverse problems, such as medical ultrasound tomography and non-destructive evaluation.

Data availability

The synthetic receiver data used in this study were generated by spectral-element simulations²⁶ and are available in the project repository at <https://github.com/x-repos/quFWI> under `data/rasht/` and `data/checkerboard/`.

Code availability

The JAX-based quantum simulator and quFWI framework presented here are available at <https://github.com/x-repos/quFWI>. Our implementation builds upon the FBPINNs framework¹⁵ (<https://github.com/benmoseley/FBPINNs>) with the PINN-based FWI formulation of Rasht-Behesht et al.¹⁴ (<https://doi.org/10.26300/x3wd-4k56>).

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Author contributions

Hoang Anh Nguyen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Divakar Vashisth:** Writing – review & editing. **Ali Tura:** Supervision, Funding acquisition, Resources, Project administration, Writing – review & editing.

Conflicts of interest

The authors declare no competing interests.

References

- [1] Albert Tarantola. Inversion of seismic reflection data in the acoustic approximation. *Geophysics*, 49(8):1259–1266, 08 1984. ISSN 0016-8033. doi: 10.1190/1.1441754. URL <https://doi.org/10.1190/1.1441754>.
- [2] Andreas Fichtner, Brian L. N. Kennett, Heiner Igel, and Hans-Peter Bunge. Full seismic waveform tomography for upper-mantle structure in the australasian region using adjoint methods. *Geophysical Journal International*, 179(3):1703–1725, 12 2009. ISSN 0956-540X. doi: 10.1111/j.1365-246X.2009.04368.x. URL <https://doi.org/10.1111/j.1365-246X.2009.04368.x>.

- [3] Ebru Bozdağ, Jeannot Trampert, and Jeroen Tromp. Misfit functions for full waveform inversion based on instantaneous phase and envelope measurements. *Geophysical Journal International*, 185(2):845–870, 05 2011. ISSN 0956-540X. doi: 10.1111/j.1365-246X.2011.04970.x. URL <https://doi.org/10.1111/j.1365-246X.2011.04970.x>.
- 440 [4] J. Virieux and S. Operto. An overview of full-waveform inversion in exploration geophysics. *Geophysics*, 74(6):WCC1–WCC26, 12 2009. ISSN 0016-8033. doi: 10.1190/1.3238367. URL <https://doi.org/10.1190/1.3238367>.
- [5] Zhongwen Zhan. Distributed acoustic sensing turns fiber-optic cables into sensitive seismic antennas. *Seismological Research Letters*, 91(1):1–15, 2020.
- 445 [6] J. Wiskin, D. T. Borup, S. A. Johnson, and M. Berggren. Non-linear inverse scattering: High resolution quantitative breast tissue tomography. *The Journal of the Acoustical Society of America*, 131(5):3802–3813, 05 2012. ISSN 0001-4966. doi: 10.1121/1.3699240. URL <https://doi.org/10.1121/1.3699240>.
- 450 [7] G Y Sandhu, C Li, O Roy, S Schmidt, and N Duric. Frequency domain ultrasound waveform tomography: breast imaging using a ring transducer. *Physics in Medicine & Biology*, 60(14):5381, jun 2015. doi: 10.1088/0031-9155/60/14/5381. URL <https://doi.org/10.1088/0031-9155/60/14/5381>.
- [8] Lihong V. Wang and Junjie Yao. A practical guide to photoacoustic tomography in the life sciences. *Nature Methods*, 13(8):627–638, 08 2016. doi: 10.1038/nmeth.3925. URL <https://doi.org/10.1038/nmeth.3925>.
- 455 [9] Jing Rao, Madis Ratassepp, and Zheng Fan. Guided wave tomography based on full waveform inversion. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, 63(5):737–745, 2016. doi: 10.1109/TUFFC.2016.2536144.
- [10] M. Raissi, P. Perdikaris, and G.E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. ISSN 0021-9991. doi: <https://doi.org/10.1016/j.jcp.2018.10.045>. URL <https://www.sciencedirect.com/science/article/pii/S0021999118307125>.
- 460 [11] Chao Song, Tariq Alkhalifah, and Umair Bin Waheed. Solving the frequency-domain acoustic vti wave equation using physics-informed neural networks. *Geophysical Journal International*, 225(2):846–859, 01 2021. ISSN 0956-540X. doi: 10.1093/gji/ggab010. URL <https://doi.org/10.1093/gji/ggab010>.
- 465 [12] Umair bin Waheed, Tariq Alkhalifah, Ehsan Haghighat, Chao Song, and Jean Virieux. Pinntomo: Seismic tomography using physics-informed neural networks, 2021. URL <https://arxiv.org/abs/2104.01588>.
- 470 [13] Xilun Liu and Mohamed Almekkawy. Full wave inversion for ultrasound tomography using physics based deep neural network. In *2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI)*, pages 1–4, 2023. doi: 10.1109/ISBI53787.2023.10230537.
- [14] Majid Rasht-Behesht, Christian Huber, Khemraj Shukla, and George Em Karniadakis. Physics-informed neural networks (pinns) for wave propagation and full waveform inversions. *Journal of Geophysical Research: Solid Earth*, 127(5), 2022. doi: <https://doi.org/10.1029/2021JB023120>. URL <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2021JB023120>.
- 475

- [15] Ben Moseley, Andrew Markham, and Tarje Nissen-Meyer. Finite basis physics-informed neural networks (FBPINNs): a scalable domain decomposition approach for solving differential equations. *Advances in Computational Mathematics*, 49(4):62, 07 2023. doi: 10.1007/s10444-023-10065-9. URL <https://doi.org/10.1007/s10444-023-10065-9>.
- [16] John Preskill. Quantum Computing in the NISQ era and beyond. *Quantum*, 2:79, August 2018. ISSN 2521-327X. doi: 10.22331/q-2018-08-06-79. URL <https://doi.org/10.22331/q-2018-08-06-79>.
- [17] Maria Schuld, Alex Bocharov, Krysta M. Svore, and Nathan Wiebe. Circuit-centric quantum classifiers. *Phys. Rev. A*, 101:032308, Mar 2020. doi: 10.1103/PhysRevA.101.032308. URL <https://link.aps.org/doi/10.1103/PhysRevA.101.032308>.
- [18] Marcello Benedetti, Erika Lloyd, Stefan Sack, and Mattia Fiorentini. Parameterized quantum circuits as machine learning models. *Quantum Science and Technology*, 4(4):043001, nov 2019. doi: 10.1088/2058-9565/ab4eb5. URL <https://doi.org/10.1088/2058-9565/ab4eb5>.
- [19] Mateusz Ostaszewski, Edward Grant, and Marcello Benedetti. Structure optimization for parameterized quantum circuits. *Quantum*, 5:391, January 2021. ISSN 2521-327X. doi: 10.22331/q-2021-01-28-391. URL <https://doi.org/10.22331/q-2021-01-28-391>.
- [20] Annie E. Paine, Vincent E. Elfving, and Oleksandr Kyriienko. Quantum quantile mechanics: Solving stochastic differential equations for generating time-series. *Advanced Quantum Technologies*, 6(10):2300065, 2023. doi: <https://doi.org/10.1002/qute.202300065>. URL <https://advanced.onlinelibrary.wiley.com/doi/abs/10.1002/qute.202300065>.
- [21] Fong Yew Leong, Wei-Bin Ewe, Si Bui Quang Tran, Zhongyuan Zhang, and Jun Yong Khoo. Hybrid quantum physics-informed neural network: Towards efficient learning of high-speed flows. *Computers & Fluids*, 301:106782, 2025. ISSN 0045-7930. doi: <https://doi.org/10.1016/j.compfluid.2025.106782>. URL <https://www.sciencedirect.com/science/article/pii/S0045793025002427>.
- [22] Stefan Berger, Norbert Hosters, and Matthias Möller. Trainable embedding quantum physics informed neural networks for solving nonlinear PDEs. *Scientific Reports*, 15(1):18823, 05 2025. doi: 10.1038/s41598-025-02959-z. URL <https://doi.org/10.1038/s41598-025-02959-z>.
- [23] Hoang Anh Nguyen and Ali Tura. Seismic traveltime inversion with quantum annealing. *Scientific Reports*, 15(1):17984, 2025. ISSN 2045-2322. doi: 10.1038/s41598-025-01188-8. URL <https://doi.org/10.1038/s41598-025-01188-8>.
- [24] Divakar Vashisth, Rodney Lessard, and Tapan Mukerji. Pre-stack and post-stack seismic inversion using quantum computing. *Geophysics*, 04 2026. doi: 10.1190/GEO-2025-0186. URL <https://doi.org/10.1190/GEO-2025-0186>.
- [25] Divakar Vashisth, Rohan Sharma, Tejas Ganesh Iyer, Tapan Mukerji, and Mrinal K. Sen. Seismic inversion using hybrid quantum neural networks, 2025. URL <https://arxiv.org/abs/2503.05009>.
- [26] Dimitri Komatitsch and Jean-Pierre Vilotte. The spectral element method: An efficient tool to simulate the seismic response of 2d and 3d geological structures. *Bulletin of the Seismological Society of America*, 88(2):368–392, 04 1998. ISSN 0037-1106. doi: 10.1785/BSSA0880020368. URL <https://doi.org/10.1785/BSSA0880020368>.
- [27] M. Cerezo, Andrew Arrasmith, Ryan Babbush, Simon C. Benjamin, Suguru Endo, Keisuke Fujii, Jarrod R. McClean, Kosuke Mitarai, Xiao Yuan, Lukasz Cincio, and Patrick J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3(9):625–644, 2021.

ISSN 2522-5820. doi: 10.1038/s42254-021-00348-9. URL <https://doi.org/10.1038/s42254-021-00348-9>.

- 525 [28] Maria Schuld, Ryan Sweke, and Johannes Jakob Meyer. Effect of data encoding on the expressive power of variational quantum-machine-learning models. *Phys. Rev. A*, 103: 032430, Mar 2021. doi: 10.1103/PhysRevA.103.032430. URL <https://link.aps.org/doi/10.1103/PhysRevA.103.032430>.
- [29] Matthew Tancik, Pratul P. Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan, 530 Utkarsh Singhal, Ravi Ramamoorthi, Jonathan T. Barron, and Ren Ng. Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems*, volume 33, pages 7537–7547, 2020.
- [30] Vincent Sitzmann, Julien N. P. Martel, Alexander W. Bergman, David B. Lindell, and Gordon Wetzstein. Implicit neural representations with periodic activation functions. In 535 *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS ’20, Red Hook, NY, USA, 2020. Curran Associates Inc. ISBN 9781713829546.
- [31] Jarrod R. McClean, Sergio Boixo, Vadim N. Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9 (1):4812, 2018. ISSN 2041-1723. doi: 10.1038/s41467-018-07090-4. URL <https://doi.org/10.1038/s41467-018-07090-4>. 540
- [32] Aditi S. Krishnapriyan, Amir Gholami, Shandian Zhe, Robert M. Kirby, and Michael W. Mahoney. Characterizing possible failure modes in physics-informed neural networks. In *Proceedings of the 35th International Conference on Neural Information Processing Systems*, NIPS ’21, Red Hook, NY, USA, 2021. Curran Associates Inc. ISBN 9781713845393.
- 545 [33] Sukin Sim, Peter D. Johnson, and Alán Aspuru-Guzik. Expressibility and entangling capability of parameterized quantum circuits for hybrid quantum-classical algorithms. *Advanced Quantum Technologies*, 2(12):1900070, 2019. doi: 10.1002/qute.201900070. URL <https://advanced.onlinelibrary.wiley.com/doi/abs/10.1002/qute.201900070>.
- [34] Maria Schuld and Francesco Petruccione. *Machine Learning with Quantum Computers*. 550 Quantum Science and Technology. Springer, 2nd edition, 2021.
- [35] James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas, Skye Wanderman-Milne, and Qiao Zhang. JAX: composable transformations of Python+NumPy programs, 2018. URL <http://github.com/jax-ml/jax>.
- 555 [36] Ville Bergholm, Josh Izaac, Maria Schuld, Christian Gogolin, Shahnawaz Ahmed, Vishnu Ajith, M. Sohaib Alam, Guillermo Alonso-Linaje, B. AkashNarayanan, Ali Asadi, Juan Miguel Arrazola, Utkarsh Azad, Sam Banning, Carsten Blank, Thomas R Bromley, Benjamin A. Cordier, Jack Ceroni, Alain Delgado, Olivia Di Matteo, Amintor Dusko, Tanya Garg, Diego Guala, Anthony Hayes, Ryan Hill, Aroosa Ijaz, Theodor Isacsson, David Ittah, 560 Soran Jahangiri, Prateek Jain, Edward Jiang, Ankit Khandelwal, Korbinian Kottmann, Robert A. Lang, Christina Lee, Thomas Loke, Angus Lowe, Keri McKiernan, Johannes Jakob Meyer, J. A. Montañez-Barrera, Romain Moyard, Zeyue Niu, Lee James O’Riordan, Steven Oud, Ashish Panigrahi, Chae-Yeun Park, Daniel Polatajko, Nicolás Quesada, Chase Roberts, Nahum Sá, Isidor Schoch, Borun Shi, Shuli Shu, Sukin Sim, Arshpreet Singh, Ingrid Strandberg, Jay Soni, Antal Száva, Slimane Thabet, Rodrigo A. Vargas-Hernández, Trevor Vincent, 565 Nicola Vitucci, Maurice Weber, David Wierichs, Roeland Wiersema, Moritz Willmann, Vincent Wong, Shaoming Zhang, and Nathan Killoran. PennyLane: Automatic differentiation of hybrid quantum-classical computations, 2022. URL <https://arxiv.org/abs/1811.04968>.

- [37] Ali Javadi-Abhari, Matthew Treinish, Kevin Krsulich, Christopher J. Wood, Jake Lishman, Julien Gacon, Simon Martiel, Paul D. Nation, Lev S. Bishop, Andrew W. Cross, Blake R. Johnson, and Jay M. Gambetta. Quantum computing with qiskit, 2024. URL <https://arxiv.org/abs/2405.08810>.
- [38] Kosuke Mitarai, Makoto Negoro, Masahiro Kitagawa, and Keisuke Fujii. Quantum circuit learning. *Physical Review A*, 98(3):032309, Sep 2018. doi: 10.1103/PhysRevA.98.032309. URL <https://link.aps.org/doi/10.1103/PhysRevA.98.032309>.
- [39] Maria Schuld, Ville Bergholm, Christian Gogolin, Josh Izaac, and Nathan Killoran. Evaluating analytic gradients on quantum hardware. *Physical Review A*, 99(3):032331, Mar 2019. doi: 10.1103/PhysRevA.99.032331. URL <https://link.aps.org/doi/10.1103/PhysRevA.99.032331>.